

Exploratory Data Analysis (EDA) Report

Task 3: Exploratory Data Analysis

1. Introduction

Exploratory Data Analysis (EDA) is the process of analyzing and visualizing data to understand its structure, patterns, relationships, and trends. EDA helps in gaining insights before performing advanced analysis or machine learning.

2. Dataset Information

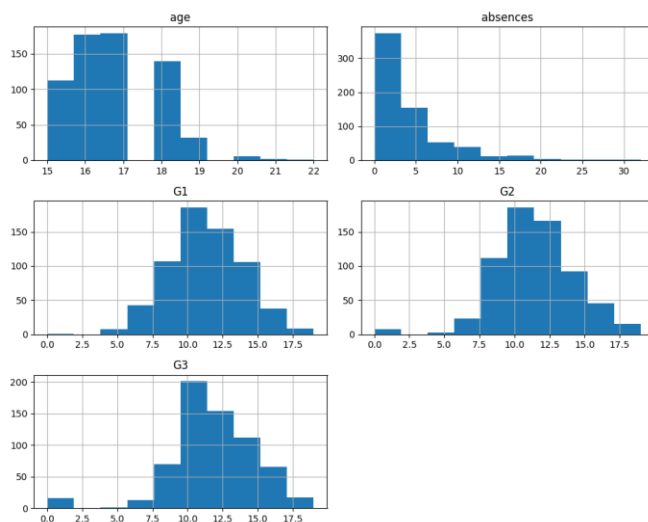
- Dataset Name: Student Performance Dataset (Portuguese Language)
- Source: Kaggle
- Dataset Link: <https://www.kaggle.com/datasets/larsen0966/student-performance-data-set>
- Number of Records: 649
- Number of Features: 33
- File Format: CSV

3. Summary Statistics

The dataset contains information related to students' demographic details, family background, study habits, and academic performance. Summary statistics were generated using Pandas to understand the distribution of numerical features.

4. Visual Analysis

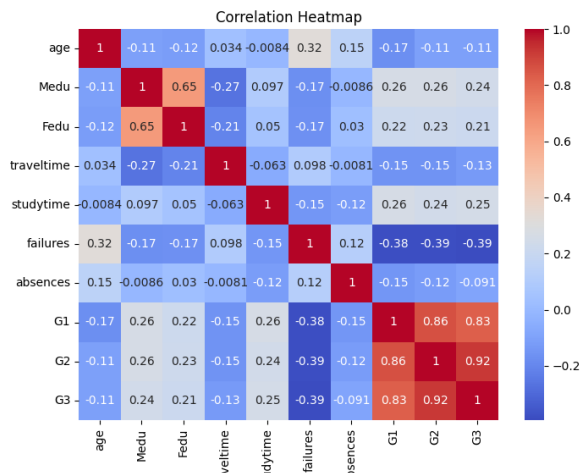
A. Histogram



Observation:

- Most students belong to a similar age group.
- Student grades are concentrated around the middle range.
- Extreme values are relatively few.

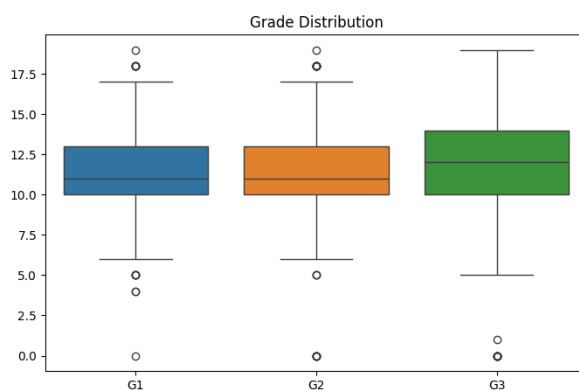
B. Correlation Heatmap



Observation:

- G1, G2, and G3 grades have strong positive correlations.
- Study time shows a positive relationship with student performance.
- Previous failures tend to negatively affect grades.

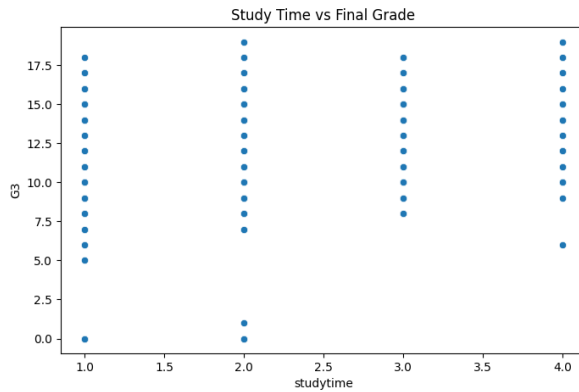
C. Boxplot



Observation:

- Most students have grades within a normal range.
- A few outliers are present, indicating exceptionally high or low performance.

D. Study Time vs Final Grade



Observation:

- Students with higher study time generally tend to achieve better grades.
- Variation exists within each study-time category.

5. Key Findings

- Academic grades (G1, G2, and G3) are strongly correlated.
- Study habits influence final performance.
- A few outliers are present in the dataset.
- Correlation analysis helps identify important factors affecting student achievement.

6. Conclusion

EDA helped in understanding the structure and characteristics of the Student Performance Dataset. The analysis identified relationships between study habits and grades, highlighted important correlations among features, and provided useful insights for future analysis and predictive modeling.